

National University of Computer and Emerging Sciences



Data Warehousing and Business Intelligence

Final Project Proposal

Financial Market and Cryptocurrency Data Warehouse

Cryptonauts

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Introduction

Our project is about building a **Financial Market & Cryptocurrency Data Warehouse**. We will combine stock market data, cryptocurrency prices, and user trading activity into one system. Using a **star schema**, the warehouse will store historical data in fact and dimension tables.

The goal is to analyze **price trends, portfolio performance, market risk, and regional trading behavior**. We will use ETL processes to clean and load the data, and BI tools to create dashboards and KPIs. This project will help us apply all concepts of DW & BI in a practical way.

1. Chosen Domain / Dataset

Our project domain is **Financial Market & Cryptocurrency Analytics**.

We have downloaded two Kaggle CSV datasets:

- **Cryptocurrency Historical Prices** – includes daily OHLC (Open, High, Low, Close) and volume data for multiple cryptocurrencies such as Bitcoin and Ethereum.
- **US Stock Market History Data** – includes daily OHLC and volume data for U.S. stocks.

In addition, we will generate **synthetic CSVs** for user trades, user information, and exchange/region metadata to complete the schema.

2. Expected Fact and Dimension Tables (Star Schema)

Fact Tables

- **FactMarketPrices** (PriceID, AssetKey, DateKey, OpenPrice, ClosePrice, HighPrice, LowPrice, Volume)
- **FactTrading** (TradeID, UserKey, AssetKey, DateKey, ExchangeKey, Quantity, TradePrice, TradeValue)

Dimension Tables

- **DimAsset** (AssetKey, AssetID, AssetName, AssetType [Stock/Crypto], Symbol, Sector/Category)
- **DimDate** (DateKey, Date, Day, Month, Quarter, Year)
- **DimUser** (UserKey, UserID, Name, Region, AccountType, SignupDate)
- **DimExchange** (ExchangeKey, ExchangeName, MarketType, Country)
- **DimRegion** (RegionKey, Country, Continent, TimeZone)

3. Planned ETL/ELT Workflow

- **Extract:** Import CSVs for stock prices, crypto prices, synthetic trades, users, and exchanges into staging tables.
- **Transform:**
 - Standardize date/time formats.
 - Normalize asset symbols across stock and crypto datasets.
 - Create surrogate keys for all dimensions.
 - Handle missing values and duplicates.
 - Derive calculated measures such as daily returns and volatility.
- **Load:** Populate the star schema into the DW schema for reporting and BI.

4. Plan for Analytics & BI Outputs

We will design dashboards and KPIs (in Power BI/Tableau) for:

- **Price Trends:** Comparison of stock vs. cryptocurrency movements over time.
- **Portfolio Performance:** User-level trading gains/losses and ROI.
- **Risk & Correlation:** Volatility metrics, correlations between crypto and stock markets.
- **Regional Behavior:** Trading activity by exchange and geographic region.
- **Market Insights:** Top performing assets, highest volume trading days, and user activity distribution.